

Passive Optical Networks (PON)

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Text Books:

1- Gerd Keiser, "FTTX concepts and applications", A John Wiley & Sons Inc., 2006

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Introduction

- In its most general form, an optical network will contain both active and passive optical elements. Active components can be located at the central office, within termination points at the customer's premises, and in the repeaters, switches, and other equipment located in the transmission path between the central office and the customer.
- Many of these devices are capable of being reconfigured by either a local or a remote-control mechanism. They are used for functions such as redirecting the light signal to another transmission path, splitting the signal into two or more branches, amplifying the optical signal power, and processing information contained in the signal.



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Introduction

- If active devices are used in the transmission path, they require electrical power to perform their functions. In addition, some type of dynamic status monitoring mechanism is needed to verify that the active devices are working properly.
- If the status information indicates a pending or actual malfunction, a network operator needs to take action and dispatch a maintenance person to the fielded unit to correct the fault. These factors add cost and complexity to the system operation compared to the case in which no active devices are used between network endpoints.



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- In a PON, there are no active components at any intermediate point between the central office and the customer's premises. Instead, only passive optical components are placed in the network transmission path to guide the traffic signals contained within specific optical wavelengths to the user endpoints and back to the central office.
- Replacing active devices with passive components provides a cost savings to the service provider
- In addition, since the passive devices have no electrical power or signal-processing requirements, they have virtually an unlimited *mean time between failures* (MTBF). This obviously lowers the overall maintenance costs significantly for the service provider.

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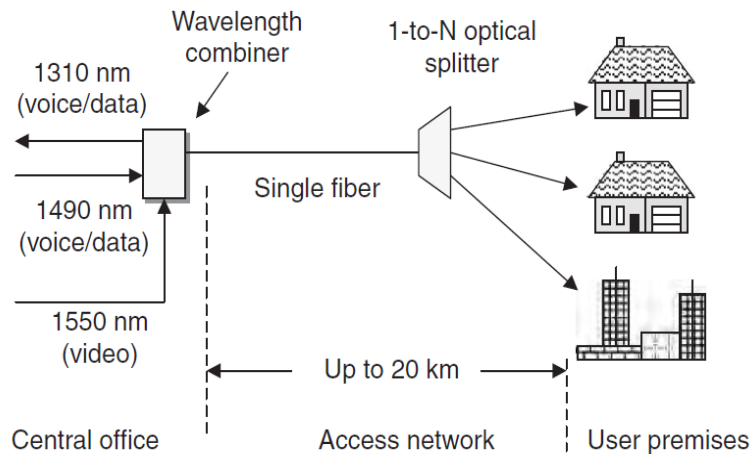
PON

- In a typical PON, a fiber optic network connects switching equipment in a central office with a number of service subscribers.
- In the central office, data and digitized voice are combined and sent downstream to customers over an optical link using a 1490-nm wavelength.
- The upstream (customer to central office) return path for the data and voice uses a 1310-nm wavelength.
- Video services are sent downstream with a 1550-nm wavelength. There is no video service in the upstream direction.

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Architecture of a typical passive optical network.

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- A **multiple dwelling unit (MDU)** refers to apartment complexes, or a building with multiple premises.
- A **multiple tenant unit (MTU)** designates an office building, an office campus, or an industrial campus with different business tenants.
- A **multiple hospitality unit (MHU)** refers to premises such as hotels, hospitals, airports, or convention centers.
- A **single dwelling unit (SDU)** is a single premises occupied by a single family or a building with two premises on one lot.

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- The transmission equipment in the network consists of an *optical line terminal* (OLT) situated at the central office and an *optical network terminal* (ONT) at each customer's premises.
- Starting at the central office, one single-mode optical fiber length connects to a passive *optical power splitter* near a housing complex, or some other campus environment.
- At this point the splitting device simply divides the optical power into N separate paths to the subscribers.



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- If the splitter is designed to divide the incident optical power evenly, and if P is the optical power entering the splitter, the power level going to each subscriber is P/N . Designs of power dividers with other splitting ratios are also possible, and there could be more than one splitter in a particular path, depending on the application.
- The number of splitting paths can vary from 2 to 64, but typically, is 8, 16, or 32. From the optical splitter, individual single-mode fibers then run to each building or to the serving equipment.
- The optical fiber transmission span from the central office to the user can be up to 20 km. Thus, active devices exist only in the central office and at the user end terminal.



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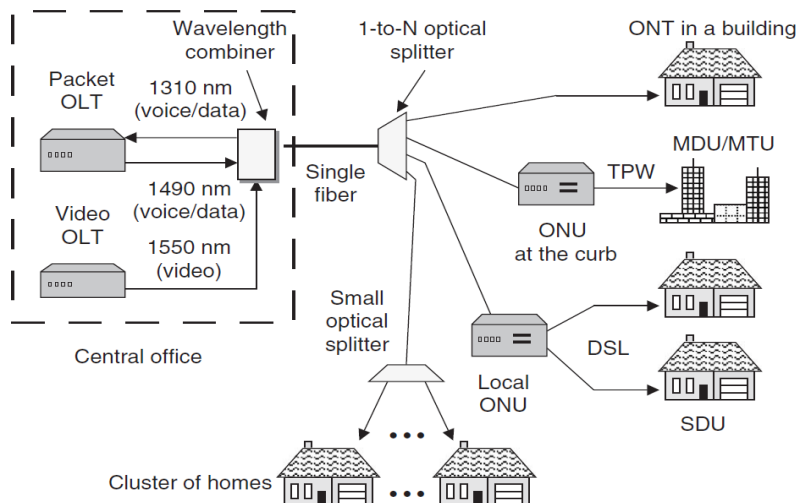
- An optical network terminal (**ONT**) is used when the fiber extends into the customer premises, whereas an optical network unit (**ONU**) is used when the fiber line terminates in some type of telecommunication cabinet located near a cluster of homes or businesses.
- Connections from the ONU to the premises can be by means of media such as twisted-pair wires (e.g., telephone lines or digital subscriber links) or coaxial cable.



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Fundamental building blocks of a PON.



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- For some situations it may be advantageous from a cost-savings point of view to run a single fiber line from the main optical splitter to a distant localized cluster of homes or small businesses or within a centralized location in a neighborhood.
- In this case, a small optical splitter is located at the end of this fiber line and then short links run from this point to the individual user premises. In contrast to running a long fiber link from the main splitter to each remote customer, this layout lowers the overall network deployment costs significantly.
- The term *optical distribution network* (ODN) refers to the collection of fibers and passive optical splitters or couplers that lie between the OLT and the various ONTs and ONUs.

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Active PON modules

Optical Line Terminal (OLT)

- The OLT is located in a central office and controls the bidirectional flow of information across the ODN.
- In the downstream direction the function of an OLT is to take in voice, data, and video traffic from a long-haul or metro network and broadcast it to all the ONT modules on the ODN.
- In the reverse direction (upstream), an OLT accepts and distributes multiple types of voice and data traffic from the network users.

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Active PON modules

Optical Line Terminal (OLT)

- A typical OLT is designed to control more than one PON. Following figure gives an example of an OLT that is capable of serving four independent passive optical networks. In this case, if there are 32 connections to each PON, the OLT can distribute information to 128 ONTs.



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Active PON modules

Optical Line Terminal (OLT)

- Passive WDM couplers (MUX) perform the wavelength combining and separation functions.
- Depending on the particular PON standard being used, the downstream and upstream transmission equipment operates at 155 Mbps, 622 Mbps, 1.25 Gbps, or 2.5 Gbps.
- In some cases the transmission rates are the same in either direction (a *symmetric* network). In other PON standards the downstream rate may be higher than the upstream rate, which is called an *asymmetric* implementation.

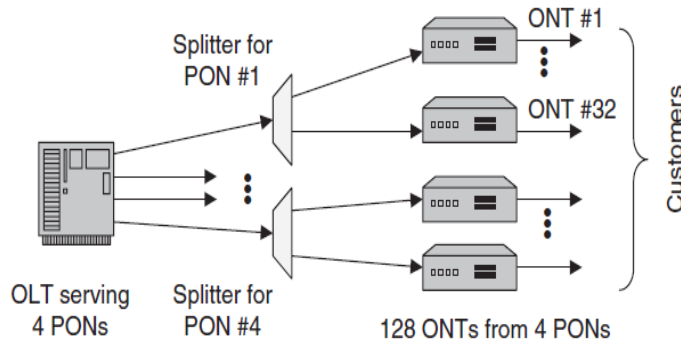


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Active PON modules

Optical Line Terminal (OLT)



OLT that is capable of serving four independent passive optical networks.

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Active PON modules

Optical Network Terminal (ONT)/

Optical Network Unit (ONU)

- An ONT/ONU is located directly at the customer's premises. Its purpose is to provide an optical connection to the PON on the upstream side and to interface electrically to the customer equipment on the other side.
- At the high-performance end, an ONT can aggregate, groom, and transport various types of information traffic coming from the user site and send it upstream over a single-fiber PON infrastructure.
- In conjunction with the OLT, an ONT allows dynamic bandwidth allocation.

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What is FTTx?

- The application of PON technology for providing broadband connectivity in the access network to homes, multiple-occupancy units, and small businesses commonly is called *fiber-to-the-x*. Here x is a letter indicating how close the fiber endpoint comes to the actual user.

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FTTx

Among the acronyms used in the technical and commercial literature are the following:

- **FTTB, *fiber-to-the-business***, refers to the deployment of optical fiber from a central office switch directly into an enterprise.
- **FTTC, *fiber-to-the-curb***, describes running optical fiber cables from central office equipment to a communication switch located within about 300 m of a home or enterprise. Coaxial cable, twisted-pair copper wires (e.g., for DSL), or some other transmission medium is used to connect the curbside equipment to customers in a building.
- **FTTH, *fiber-to-the-home***, refers to the deployment of optical fiber from a central office switch directly into a home. The difference between FTTB and FTTH is that typically, businesses demand larger bandwidths over a greater part of the day than do home users.

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FTTx

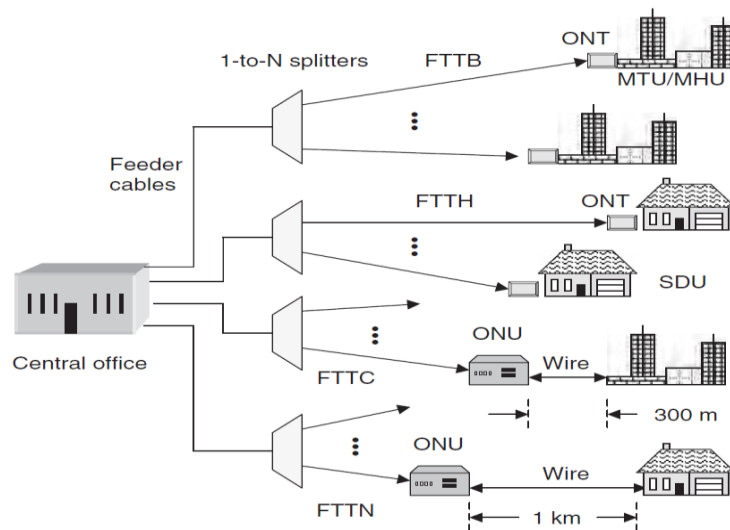
- **FTTN**, *fiber-to-the-neighborhood*, refers to a PON architecture in which optical fiber cables run to within about 1 km of homes and businesses being served by the network.
- **FTTO**, *fiber-to-the-office*, is analogous to FTTB in that an optical path is provided all the way to the premises of a business customer.
- **FTTP**, *fiber-to-the-premises*, has become the prevailing term that encompasses the various FTTx concepts. Thus FTTP architectures include FTTB and FTTH implementations.

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FTTx



Some FTTx scenarios.

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Downstream

Traffic Flows

- The OLT uses *time-division multiplexing* (TDM) to combine incoming voice and data streams that are destined for users on the PON.
- The multiplexed downstream signal is broadcast to all the ONTs. Each ONT discards or accepts the incoming information packets, depending on the packet header addressing (logical id).
- Encryption may be necessary to maintain privacy, since the downstream signal is broadcast and every ONT receives all the information destined for each end terminal.

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Encryption for downstream flow

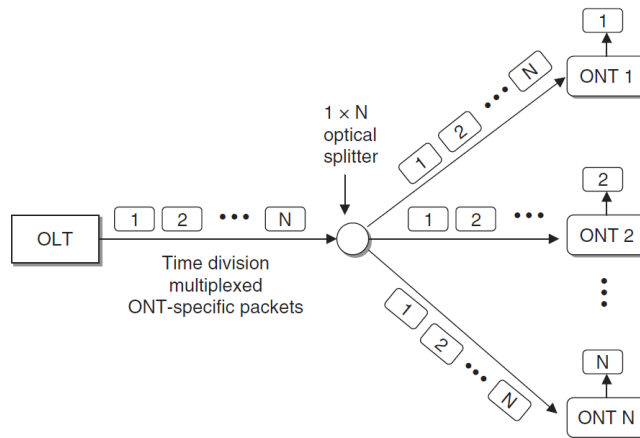
- OLT initiates a key exchange request to the ONU. ONU responds to the request with a new key.
- After receipt of the key, the OLT uses the new key to encrypt data.
- OLT sends the encrypted data to the ONU.
- ONU receives the encrypted data and uses the key to decrypt it.

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Traffic Flows



Operation of a time-division multiplexing process.

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Upstream

Traffic Flows

- One important function of an OLT is to assign bandwidth dynamically to the ONT modules.
- Since up to 64 ONTs use the same wavelength and share a common optical fiber transmission line, some type of transmission synchronization must be used to avoid collisions between traffic coming from different ONTs in the upstream direction.
- The simplest method is to use *time-division multiple access* (TDMA), wherein each user transmits information within a specific assigned time slot at a prearranged data rate.

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Upstream

Traffic Flows

- The OLT controls and coordinates the traffic from each ONT by sending permissions to them to transmit during a specific time slot. The time slots are synchronized so that transmission bursts from different users do not collide.
- However, this does not make efficient use of the bandwidth available since many time slots will be empty when several network users do not have information to be sent back to the central office.
- A more efficient process is *dynamic bandwidth allocation (DBA)*, wherein time slots of an idle or low-utilization user are assigned to a more active customer.

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Upstream

Dynamic Bandwidth Allocation (DBA)

- Technique by which traffic bandwidth is allocated on demand and fairly between different users in a shared telecommunications medium. The sharing of a link adapts in some way to the instantaneous traffic demands of the nodes connected to the link.
- It makes use of the fact that users are not transmitting data at all times.

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Upstream

Traffic Flows

- Since each end terminal is located at different distances from the central office, the OLT uses a **Ranging** technique to measure the logical distance between the users and the OLT. This ranging technique calculate the round trip delay **RTD** to measure distances. This enables each ONT to adjust its transmission timing properly to avoid traffic collisions.

1-Distance between the OLT and ONT/ONU is measured.

2-Time slots are allocated based on distance.

3-ONT/ONU sends traffic upstream based on granted time slot.

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Upstream

Traffic Flows

- To resolve the upstream dynamic bandwidth allocation problem, a handshaking procedure between an ONU and an OLT is defined.
- Two control messages are used: the **report** message, which contains the number of time slots an ONU needs, (sent by an ONU to an OLT), and the **gate** message, which contains a grant indicating the start time of frame transmission and the duration,(sent by an OLT to an ONU).

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Power Budgeting

- As part of GPON, loss of optical power must be accounted for. This loss can be introduced in a variety of ways, such as:
 - Loss within splitters
 - Loss per km of fiber (About 0.35 dB per km for 1310, 1490 nm)
 - Loss in splices (> 0.2 dB)
 - Connector loss (~ 0.6 dB)
 - Fiber bending



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Power Budgeting

- Approximate loss incurred from use of 1 x 2 splitter = 3.5 dB:
- Approximate loss incurred from use of various splitters:

Optical Splitters	Loss [dB]
Splitter 1 x 64	20.1
Splitter 1 x 32	17.4
Splitter 1 x 16	13.8
Splitter 1 x 8	10.5
Splitter 1 x 4	7.0



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Passive components for PON

Optical power splitter structure

- High-quality optical splitters can be made by using either *planar integrated optical circuit* technology or a series of 2x2 fused-fiber single-mode couplers.
- Following figures illustrate the structural concepts of using fused-fiber and planar integrated optical splitters, respectively, for making a 1x8 optical power splitter.
- When constructing a splitter using 2x2 fused-fiber couplers, only one input arm is used.



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Passive components for PON

Optical power splitter structure

- The integrated optical splitter consists of a series of joined Y-coupler waveguides.
- The operation of these devices is reciprocal; that is, they can function as either optical power splitters to divide an incoming light signal into $N \leq 32$ branches (nominally for a PON) or as power-combining devices to multiplex optical signal streams from individual fibers onto a common fiber.
- The planar integrated optical splitters offer more compact packages than fused-fiber splitters for situations where size is an issue.

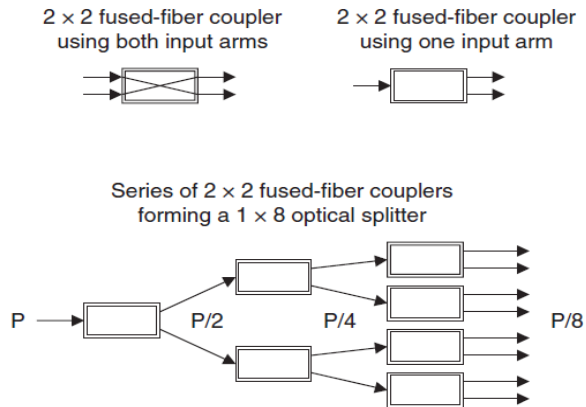


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Passive components for PON

Optical power splitter structure



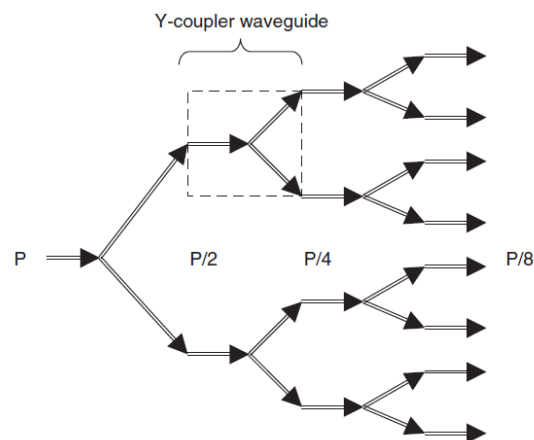
Structural concepts of using fused-fiber couplers for making a 1 x 8 optical power



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Passive components for PON

Optical power splitter structure



Structural concepts of using PLC splitters for making a 1 x 8 optical power splitter.



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PON alternatives

- There are several alternative PON implementation schemes, the two main ones being Ethernet PON (**EPON**), and gigabit PON (**GPON**). The key differences between them lie in the transmission protocols that are employed.
- While EPON employ Ethernet protocol, GPON employ ATM and Ethernet. Following table lists some of the characteristics of each methodology and the standards to which they adhere.

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EPON & GPON

	EPON	GPON
Standards Family	IEEE 802.3ah	ITU-T G.984
Protocol	Ethernet	ATM and Ethernet
Transmission speeds (Mbps)	1244 upstream, 1244 downstream	622 or 1244 upstream, 1244 or 2488 downstream
Span	10 or 20 km	10 or 20 km
Number of splits	16 : 32	32 Up to (128)

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